



UNIVERSIDAD TECNOLÓGICA DE PANAMÁ
CENTRO REGIONAL DE COCLÉ
FACULTAD DE INGENIERÍA DE SISTEMAS COMPUTACIONALES
LICENCIATURA EN CIBERSEGURIDAD

BASE DE DATOS SEGURAS

PRÁCTICA #8

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- Crear trigger para borrar empleados

```
65     DELIMITER $$
66
67 •   CREATE TRIGGER trg_delete_empleados
68
69     AFTER DELETE ON empleados
70
71     FOR EACH ROW
72
73     BEGIN
74
75     INSERT INTO auditoria_eventos
76     (
77     usuario,
78     accion,
79     tabla_afectada,
80     fecha_evento
81     )
82
83     VALUES
84     (
85     USER(),
86     'DELETE',
87     'empleados',
88     NOW()
89     );
90
91     END $$
92
93     DELIMITER ;
```

- Trigger insertar empleado

```
3      DELIMITER $$
4
5  •   CREATE TRIGGER trg_insert_empleados
6
7      AFTER INSERT ON empleados
8
9      FOR EACH ROW
10
11  BEGIN
12
13      INSERT INTO auditoria_eventos
14  (
15      usuario,
16      accion,
17      tabla_afectada,
18      fecha_evento
19  )
20
21      VALUES
22  (
23      USER(),
24      'INSERT',
25      'empleados',
26      NOW()
27  );
28
29  END $$
30
31  DELIMITER ;
```

- Update empleado

```
35     DELIMITER $$
36
37 •   CREATE TRIGGER trg_update_empleados
38
39     AFTER UPDATE ON empleados
40
41     FOR EACH ROW
42
43     BEGIN
44
45     INSERT INTO auditoria_eventos
46     (
47     usuario,
48     accion,
49     tabla_afectada,
50     fecha_evento
51     )
52
53     VALUES
54     (
55     USER(),
56     'UPDATE',
57     'empleados',
58     NOW()
59     );
60
61     END $$
62
63     DELIMITER ;
```

- Conéctate como: admin_dba

```
1 • SELECT CURRENT_USER();
```

Result Grid				Filter Rows:	Export: 	Wrap Cell Content: 
	CURRENT_USER()					
	admin_dba@localhost					

```
3 • USE empresa_segura;
```

Luego:

```
DELETE FROM empleados  
WHERE id_empleado = 2;
```

```
27 • DELETE FROM empleados  
28 WHERE id_empleado = 4;
```

UPDATE

```
UPDATE empleados  
SET salario = 3200  
WHERE id_empleado = 1;
```

```
23 • UPDATE empleados  
24 SET salario = 6000  
25 WHERE id_empleado = 4;
```

SELECT *

```
FROM auditoria_eventos  
ORDER BY fecha_evento DESC;
```

	id_evento	usuario	accion	tabla_afectada	fecha_evento
▶	23	admin_dba@localhost	DELETE	empleados	2026-05-14 16:03:48
	22	admin_dba@localhost	UPDATE	empleados	2026-05-14 16:03:06
	21	root@localhost	INSERT	empleados	2026-05-14 16:01:40
	20	root@localhost	DELETE	empleados	2026-05-14 15:32:59
	19	root@localhost	UPDATE	empleados	2026-05-14 15:22:44
	18	root@localhost	INSERT	empleados	2026-05-14 15:21:34
	13	admin_dba	SELECT	empleados	2026-05-12 23:07:31
	14	admin_dba	INSERT	empleados	2026-05-12 23:07:31
	15	supervisor1	UPDATE	proyectos	2026-05-12 23:07:31
	16	analista1	SELECT	empleados	2026-05-12 23:07:31
	17	consulta1	SELECT	vista_emplead...	2026-05-12 23:07:31
*	NULL	NULL	NULL	NULL	NULL

7 • SHOW TRIGGERS;

Trigger	Event	Table	Statement	Timing	Created	sql_mode
trg_insert_empleados	INSERT	empleados	BEGIN INSERT INTO auditoria_...	AFTER	2026-05-11 17:11:00.78	ONLY_FULL
trg_update_empleados	UPDATE	empleados	BEGIN INSERT INTO auditoria_...	AFTER	2026-05-14 15:20:59.53	ONLY_FULL
trg_delete_empleados	DELETE	empleados	BEGIN INSERT INTO auditoria_...	AFTER	2026-05-14 15:28:57.82	ONLY_FULL

PRUEBA COMPLETA RECOMENDADA

```
INSERT
INSERT INTO empleados
(nombre,correo,cargo,salario,id_departamento)
```

```
VALUES
(
'Supervisor Test',
'super@test.com',
'Analista',
2500,
2
);
```

```
31 • INSERT INTO empleados
32     (nombre,correo,cargo,salario,id_departamento)
33
34     VALUES
35     (
36         'Celso Perez',
37         'celso@test.com',
38         'Fotologo',
39         4500,
40         2
41     );
```

```
UPDATE
UPDATE empleados
SET salario = 3200
WHERE id_empleado = 1;
```

```
16 • UPDATE empleados
17     SET salario = 5500
18     WHERE id_empleado = 29;
```

```
DELETE
DELETE FROM empleados
WHERE id_empleado = 1;
```

```
20 • DELETE FROM empleados
21     WHERE id_empleado = 29;
```

```
SELECT *  
FROM auditoria_eventos  
ORDER BY fecha_evento DESC;
```

	id_evento	usuario	accion	tabla_afectada	fecha_evento
▶	35	supervisor1@localhost	DELETE	empleados	2026-05-16 15:04:12
	34	supervisor1@localhost	UPDATE	empleados	2026-05-16 15:04:08
	33	supervisor1@localhost	INSERT	empleados	2026-05-16 15:02:56